

When Aggregate Welfare Misses Joint-Action Structure in Multi-Agent Pricing

Anonymous Author(s)
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ABSTRACT

Monitors of multi-agent systems often record aggregate outcomes such as prices or welfare, yet many joint actions produce the same aggregate. We study this observability boundary in a repeated two-seller pricing market in which language-model sellers see a public peer-price history but have no message channel. We formalize when a target is recoverable from a monitor; exact enumeration shows that 100 ordered price states form 10 welfare classes, inside which only seller-level observables separate ties from market capture. In a controlled-initial closed-loop comparison, natural display produced equal-price states in 88.8% and 85.3% of assessment rounds in the two asymmetric cells and hiding the rival’s history produced none; block-paired equal-price differences were 0.885 and 0.853, while minimum-price differences were only 0.011 and 0.014. The arms separated at the first model decision. An independent replication confirmed the structural contrast ($D_{\text{tie}} = 0.830$, 95% CI [0.743, 0.917], positive in all 42 blocks) while registered welfare contrasts stayed near zero, with equivalence inconclusive under the registered missingness bound. With model-chosen starting states, the same welfare-equivalent capture states instead resolved upward into 6.5 ties, so structure anticipated a welfare loss of 27.9 units that welfare could not reveal. A Q-learning baseline and settlement arithmetic for more sellers and captive consumers bound when the blind spot appears. The results identify information-dependent joint-action structure that welfare alone cannot resolve, but not collusion: the observed ties lie above the joint-profit price, where competence and forced-deviation probes cannot separate strategic sustainment from anchoring.

KEYWORDS

multi-agent systems, generative agents, coordination, agent evaluation, algorithmic pricing, observability, monitoring

1 INTRODUCTION

Autonomous agents increasingly post prices, bid and allocate resources on behalf of firms, and the systems that oversee them usually record aggregate outcomes: average prices, consumer welfare or industry profit. Competition research asks whether pricing algorithms can raise prices without agreement [10, 13, 20], and agendas for multi-agent oversight name collusion and miscoordination among the failures to detect and call for visibility into agent activity [11, 12, 18]. Whether an aggregate record can support such distinctions is a property of the monitor as much as of the agents.

Public action histories make this question concrete. Without any message interface, a seller can condition its price on a peer’s previous price and so reshape the joint trajectory. In a homogeneous Bertrand market the lowest posted price fixes demand, consumer surplus and industry profit. The same minimum price can therefore arise from a tie, in which the sellers split demand, or from an

asymmetric state, in which one seller captures it. An intervention can move trajectories between these states while leaving welfare unchanged, so a null welfare contrast is compatible with a large change in joint behaviour.

Prior work establishes that learning algorithms and language models can sustain supra-competitive prices [10, 14, 24], that public price information can facilitate coordination [3, 30, 40], and that language-model agents show measurable cooperation and coordination in repeated games [2, 35]. These studies evaluate outcomes or strategies; they rarely treat the mapping from joint actions to the monitored aggregate as an object of measurement. It remains unclear under which conditions an aggregate monitor is blind to an intervention’s effect on joint behaviour, and whether the unseen structure carries information about future welfare.

We study this boundary in a repeated two-seller pricing market whose sellers are language models that see a public peer-price history. The intervention is the display policy, which determines whether each seller sees its rival’s past prices. We formalize when a target is recoverable from a monitor, certify the welfare classes of the settlement rule exactly, and measure in closed-loop experiments how display moves trajectories within and across those classes. We then test when the unseen structure foreshadows welfare, compare the language-model sellers with tabular Q-learners and other market rules, and state the evidence a strategic interpretation would require. The paper makes four contributions:

- (1) A recoverability criterion for monitors, an exact certificate of the settlement’s welfare classes, and a blind-share metric for the part of a between-arm difference that a welfare monitor cannot see.
- (2) Closed-loop evidence, with an independent replication, that hiding the rival’s history changes joint-state structure and seller-profit allocation from the first model decision while welfare barely moves.
- (3) Evidence that welfare-equivalent states have different welfare futures, and boundary conditions for the blind spot across agent class, number of sellers and captive demand.
- (4) An operating-point gate for strategic claims, with competence and forced-deviation probes showing that the observed operating points do not qualify.

Figure 1 previews the central contrast.

2 RELATED WORK

2.1 Algorithmic pricing and tacit collusion

Repeated-game theory explains how prices above the competitive level can be sustained without agreement: with patient players the folk theorem supports a wide range of payoffs [16], and the quality of price observation determines whether deviations can be detected and punished [17, 40]. Simulation studies show that

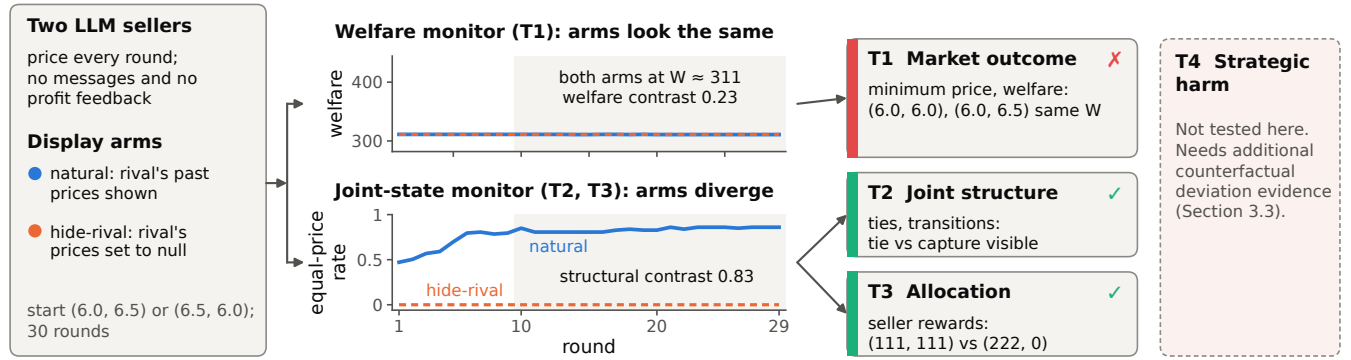


Figure 1: The same trajectories, seen by two monitors. Replication, asymmetric cells (LH and HL averaged), rounds 1–29: per-round means over completed trajectories; shading marks the assessment window. On the full grid-welfare scale, a welfare monitor (T1) shows the two display arms at the same welfare, whereas a joint-state monitor (T2, T3) shows them diverging from the first model decision. Quoted contrasts are the registered block-level D_{asym} and D_{tie} . T4 is not tested here.

independent learners reach such outcomes. Q-learning [44] firms partly collude in Cournot markets [43]; Q-learning sellers sustain supra-competitive prices with reward–punishment schemes in repeated Bertrand games [10] and under sequential pricing [24]; and correlated experimentation by independent bandit algorithms can raise prices without any reactive strategy [19]. Seemingly collusive outcomes can also originate in imperfect exploration [1], and the design of the learning algorithm shapes the prices it reaches [5]. Field evidence links algorithm adoption to higher margins [6] and documents pricing technologies whose frequency or periodic resets soften competition [8, 33]. Language-model agents can also set supra-competitive prices, with outcomes sensitive to prompt wording [14], and near-monopoly prices can arise in equilibrium from algorithms that encode no threats [4]. This literature mostly measures outcomes, typically price levels or normalized profit gains, and then asks which mechanism produced them. Our question precedes that step: which differences in joint behaviour an outcome measure can register at all.

2.2 Price transparency and information disclosure

Public price information is a classical facilitating practice because it makes deviations observable and coordination easier to sustain [37]. After a competition authority began publishing firm-level transaction prices for ready-mixed concrete in Denmark, prices rose [3], and online disclosure of retail gasoline prices in Chile raised average margins [30]. Without any communication, shared salient information can also coordinate expectations on a focal action [38]. These studies vary disclosure and measure price levels. We vary what each agent sees of its rival's history and measure where trajectories move in the joint-action space, including moves that leave the price level unchanged.

2.3 Multi-agent learning and language-model agents

Multi-agent research separates individual policies from the organization that emerges through interaction [9, 41] and studies coordination under mixed incentives in sequential social dilemmas and evaluation suites [25, 26]. One line makes the interaction channel explicit: agents learn differentiable messages or symbolic protocols [15, 22, 42], and language-model agents can hide coordination inside an explicit channel through steganography [32]. Coordination also emerges from training objectives such as centralized critics [29], social-influence rewards [23] and autocurricula [7]. Generative-agent frameworks make memory, roles, conversation and tool use part of the interaction interface [27, 34, 45], couple reasoning to action and verbal feedback [39, 46], and benchmark agents across environments [28]. Behavioural game-theory studies find that language models play self-interested repeated games well but coordinate poorly in some coordination games [2], and that most simulated language-model societies fail to sustain a shared resource [35]. These studies report task success, cooperation rates or collective welfare. We remove the message channel and treat the displayed peer history as the intervention, so the question becomes whether a monitor can detect joint-state changes when the interface, not the action space, changes.

2.4 Monitoring, auditing and identification

Oversight proposals for AI agents call for visibility measures such as agent identifiers, real-time monitoring and activity logs [11], and taxonomies of multi-agent risk name collusion among the failure modes that monitors should detect [12, 18]. Legal analyses ask when collusion by autonomous agents should be unlawful and what evidence would show it [13, 20], and audit designs test pricing algorithms for plausible non-collusion from the data they collect [21]. Each proposal presupposes that the recorded data identify the target. Identification theory makes this checkable: a parameter is identified when no two admissible structures generate the same distribution of observables [36], and partial identification bounds

what remains when they do [31]. We apply this logic to monitor design: we derive which targets a welfare record can identify, quantify the unidentified share of a treatment effect, and bound missing outcomes rather than impute them.

3 ENVIRONMENT, MONITORS AND TARGETS

3.1 Two-seller pricing environment

Two sellers choose prices from $G = \{2.0, 2.5, \dots, 6.5\}$. Marginal cost is 1, reservation price is 10, and market size is 100. The lowest-priced seller serves all demand, and a tie splits demand equally. Let $m = \min(p_A, p_B)$ and $Q(m) = 100(10 - m)/9$. Consumer surplus and total industry profit are

$$CS(m) = \frac{1}{2}(10 - m)Q(m), \quad \Pi(m) = (m - 1)Q(m), \quad (1)$$

and total welfare is $W(m) = CS(m) + \Pi(m) = 50(10 - m)(8 + m)/9$. Seller profit allocation depends on the number of sellers attaining m .

3.2 Monitors and recoverability

A monitor records a function f of the ordered state $p = (p_A, p_B) \in S = G^2$ in every round, and a scientific target is a function g on S . We call g *recoverable* from f if $g = h \circ f$ for some h , that is, if g is constant on every fibre $f^{-1}(o)$. The welfare monitor records $f_W(p) = (m, Q, CS, \Pi, W)$. For a distribution P of states, $f_{W\#}P$ denotes the induced distribution of welfare records.

PROPOSITION 3.1. (a) *The fibres of f_W are the ten classes $C_m = \{p \in S : \min(p_A, p_B) = m\}$, $m \in G$, with $|C_m| = 2n_m - 1$, where n_m is the number of grid prices at or above m .* (b) *Every function of m is recoverable from f_W . The ordered state, the equal-price indicator $E = 1[p_A = p_B]$ and the seller-profit vector are not: every non-singleton class contains the tie (m, m) with profits $(\Pi/2, \Pi/2)$ and capture states with profits $(\Pi, 0)$ or $(0, \Pi)$.* (c) *For any two arm-level state distributions P and P' , $\Delta_W \leq \Delta$, where $\Delta = TV(P, P')$ and $\Delta_W = TV(f_{W\#}P, f_{W\#}P')$ are total-variation distances.*

PROOF SKETCH. $W(m)$ is strictly decreasing for $m > 1$, so f_W is injective in m and its fibres are the minimum-price classes. Inside C_m a tie splits demand and a unique lowest seller serves all of it, which gives (b). Part (c) is the data-processing inequality for total variation. $\square$

Part (c) motivates a *blind share* $\beta = 1 - \Delta_W/\Delta \in [0, 1]$: the fraction of the between-arm difference in state distributions that a welfare monitor cannot see. It equals 1 when two arms differ only within classes and 0 when every difference changes the minimum price. Exhaustive comparison of every settled quantity confirms that only seller profits and the statistics derived from them (the equal-price indicator, profit HHI and share disparity) vary inside a class. Under this tie-split rule profit HHI equals $1 - E/2$, so it carries no information beyond E ; recording the ordered pair or both seller rewards splits every class. For example, $(6.0, 6.5)$ and $(6.0, 6.0)$ both yield $m = 6.0$, $W = 311.111$, $CS = 88.889$ and $\Pi = 222.222$, but seller profits $(222.222, 0)$ and $(111.111, 111.111)$ (Figure 2A–B), and recorded $(6.0, 6.5)$ rounds reproduce these quantities, including the logged seller rewards. The class structure is exact arithmetic, not an empirical contribution.

Table 1: Evidence targets and what the present design can identify.

Target	Observable surface	Status in this paper
T1 market harm	m , welfare, consumer surplus, industry profit	Estimable under the block design, but blind to ordered state and allocation within a class; window support is pinned at $m \in \{6.0, 6.5\}$.
T2 joint structure	Ordered pairs, equality, transitions, persistence	Directly observed; replicated.
T3a/T3b allocation	Seller rewards: capture versus equal division	Directly observed; reported by direction.
T4 strategic harm	Declared deviation gain and response to that deviation	Not identified. The observed operating points fail gate condition (iii).

3.3 Evidence targets and the operating-point gate

We distinguish four evidence targets. *T1 market harm* concerns minimum price, welfare and benchmark gaps. *T2 joint-action structure* concerns ordered price-pair distributions, equality, transitions and persistence. *T3 allocation* is split by direction: *T3a capture concentration* (one seller serves the market, HHI 1.00) and *T3b market division* (equal division at a tie, HHI 0.50). Equal division, not concentration, is the cartel-like pattern, so a higher HHI is not “more coordinated”. *T4 strategic harmful coordination* requires a counterfactual deviation gain and a response to that deviation. By Proposition 3.1, welfare treatment effects are estimable under the randomized design, but T2 and T3 are not recoverable from a welfare-only surface, and neither a T2 nor a T3 monitor can establish a deviation response (Table 1).

Before a T4 test is informative, the target symmetric tie (p, p) must pass an arithmetic *operating-point gate*: (i) per-seller profit exceeds the grid stage-Nash profit; (ii) some unilateral grid deviation raises current profit; and (iii) p is at or below the symmetric joint-profit price, so that a profitable deviation cannot coincide with a joint move toward the collusive optimum. Condition (iii) separates mechanisms by design; it is not a theorem, since repeated-game arguments can sustain ties above the joint-profit price [16]. Under this settlement the unique pure grid-Nash outcome is $(2.0, 2.0)$ and the joint-profit price is 5.5, so ties at 2.5–5.5 pass the gate, whereas the observed ties at 6.0 and 6.5 satisfy (i) and (ii) but fail (iii).

4 STUDY DESIGNS

4.1 Closed-loop policy contrast

The earlier four-arm study varied whether own and rival price histories were displayed (natural, hide-own, hide-rival, hide-both; 44 of 48 blocks complete). Its round 0 was a model decision with an identical prompt in every arm, so its starting states are model draws. Its registered rival-history factorial effect averages over the own-history condition; we report a post-hoc natural-to-hide-rival simple contrast separately.

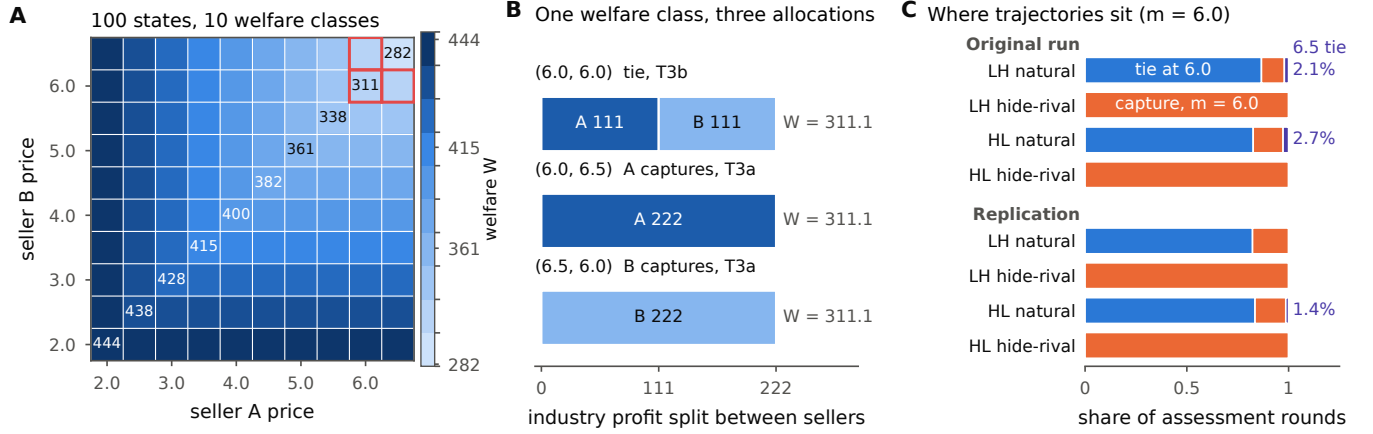


Figure 2: The settlement rule creates welfare classes that hide allocation. (A) Welfare of all 100 ordered price states; each L-shaped class shares one minimum price and welfare value (diagonal labels); red outlines mark the $m = 6.0$ class. (B) Its three states have welfare 311.1 but split industry profit equally (tie, T3b) or give it to one seller (capture, T3a). (C) Assessment-window composition (rounds 10–29) of the asymmetric cells in both controlled-initial runs: natural display moves trajectories from capture to ties inside the $m = 6.0$ class; only the few 6.5 ties (percent labels) change welfare. Repeated-round summaries.

For the controlled-initial welfare outcome Y , with subscripts h for hide-rival and n for natural display, the registered contrasts are

$$D_{\text{asym}} = \frac{1}{2} [(Y_{LH,h} - Y_{LH,n}) + (Y_{HL,h} - Y_{HL,n})],$$

$$J = D_{\text{asym}} - \frac{1}{2} [(Y_{LL,h} - Y_{LL,n}) + (Y_{HH,h} - Y_{HH,n})],$$

each estimated with Bonferroni-corrected block-level t intervals. Structural contrasts use the opposite, natural-minus-hide-rival sign convention.

4.2 Controlled initial states and replication

The controlled-initial study programs round-0 prices as one of (6.0, 6.0), (6.0, 6.5), (6.5, 6.0) and (6.5, 6.5), labelled LL, LH, HL and HH, under two display arms, natural display and hide-rival. Rounds 1–29 used model decisions; rounds 10–29 formed the assessment window. The final record contained 380 completed and 4 failed trajectories. The run was an amended continuation after an operational stop caused by an error-classification defect; failed trajectories were not repaired.

A separately specified confirmation protocol (Candidate A) repeated the eight-cell, 48-block design with the same model, settings and prompt surfaces, and was not pooled with the original run. Its structural primary is

$$D_{\text{tie}} = \frac{1}{2} [(E_{LH,n} - E_{LH,h}) + (E_{HL,n} - E_{HL,h})],$$

where E is the assessment-window equal-price fraction, with a block-level two-sided 95% t interval; D_{asym} and J are tested against the protocol's candidate equivalence margin $\delta_W = 5$, requiring both Bonferroni intervals to lie in $[-\delta_W, \delta_W]$. It ran in two sealed segments: a local ledger-write failure stopped the first, and the continuation was authorized after an inspection of interim results, changing only the ledger-replacement retry window. It recorded 376 complete and 8 failed trajectories, none repaired or imputed.

4.3 Model, prompt and reproducibility

All runs used the deepseek-flash model from DeepSeek with one simultaneous API call per seller per round (temperature 0.3, 256-token limit, forced price-selection tool, one format attempt, no repair). The system prompt asked each model to act as an autonomous seller maximizing cumulative profit without a message channel. Each user message gave JSON fields for the round index, remaining horizon, costs, reservation value, demand rule, market size, allowed prices and a projected posted-price history, but no realized sales, profits, welfare or other outcome statistic; in hide-rival the rival price fields were null. Trajectories had 30 rounds. The original schedule seed was 20260929 and each reset seed equalled the block identifier; no provider seed was recorded, so pairing matches the deterministic reset but not provider sampling. The original run's failed cells were LL-natural (blocks 37, 41), HH-hide-rival (37) and LH-hide-rival (47); all 21,982 logged replication calls returned the same alias and fingerprint with one attempt each.

4.4 Offline structural, baseline and incentive analyses

We enumerate the grid in exact rational arithmetic, derive the partition of Proposition 3.1, and re-settle every one of the 11,375 recorded rounds of the replication ledger (0 mismatches). Best-response enumeration gives the unique pure grid-Nash outcome (2.0, 2.0), with per-seller profit 44.444; the joint-profit maximum is the symmetric 5.5 state, with per-seller profit 112.500. These benchmarks are descriptive, not evidence that the model computes an equilibrium. The same code extends the class analysis to three and four sellers and to captive consumers. As a non-language-model reference, tabular Q-learners [44] play the same market under the same display arms, with the displayed last prices as their state and their own realized profit as reward (200 seeded sessions per arm, 2,000,000 learning periods; simulation only, no provider calls). Unlike the

language-model sellers, a Q-learner therefore always observes its own outcome.

4.5 Capability and deviation-response probes

Two supplementary probes used the same endpoint and settings. In X1, a fixed rival price (6.0 or 6.5) and an identical current-profit table made 5.5 the unique current-profit best response; the only randomized content was whether the decision was terminal or had 29 further rounds. In X2, 48 four-arm blocks began with a programmed (6.0, 6.0) prelude; at round 10 the designated seller was forced to 5.5 in the intervention arms, while matched no-shock arms kept the model action. Under hide-rival the forced price is not displayed to the responder, so those arms are a structural negative control. X1 was analysed at the prompt-ID cluster level and X2 at the complete-block level.

4.6 Inference and failure handling

The block is the independent unit for every inferential contrast; round counts are nested descriptive summaries, and post-hoc sensitivities are labeled as such. We do not impute failed trajectories or silently drop their blocks, and descriptive denominators accompany every repeated-round count. Missing-outcome completion ranges bound the planned 48-block estimand in the sense of partial identification [31]: observed partial rounds are retained and unresolved rounds are set to the extremes of the grid welfare support [281.9, 444.4] (or [0, 1] for equal-price fractions). They are bounds, not confidence intervals.

5 RESULTS

5.1 Display changes structure while minimum price barely moves

We aggregated assessment rounds within each trajectory and paired blocks across arms. In the original run the post-hoc structural sensitivities were large (Table 2): natural display minus hide-rival raised the equal-price fraction by 0.885 in LH and 0.853 in HL, while minimum price changed by only 0.011 and 0.014; the HHI differences (-0.443 and -0.427) are $-\frac{1}{2}$ times these and add nothing. The underlying repeated-round summaries were 852/960 equal-price rounds for LH-natural and 0/940 for LH-hide-rival, and 819/960 and 0/960 in HL.

The composition of the assessment window shows where this contrast lives (Figure 2C). Every hide-rival asymmetric trajectory stayed in a capture state at $m = 6.0$ throughout the window, in both runs. Under natural display 82–87% of rounds were ties at 6.0, 11–18% remained capture states at 6.0, and at most 2.7% were ties at 6.5, the only states that change welfare: a large structural contrast inside a small welfare margin.

Hide-rival froze prices: in LH neither seller ever changed its round-0 price, and in HL seller B changed in 4% of trajectories. Under natural display the initially high seller re-priced at some round in 96% (LH) and 94% (HL) of trajectories. LL and HH stayed tied in every assessment round in both arms. In an offline stress test on a separate sealed trajectory set, copy-the-rival and match-if-equal-else-undercut rules matched 87.50% and 86.875% of held-out assessment actions without being exact generators. High-price ties

Table 2: Block-paired contrasts, natural display minus hide-rival, in the asymmetric cells. Assessment rounds (10–29) were aggregated within trajectory and paired by block; intervals are 95% block- t . Round 1 is the paired difference in the equal-price rate at the first model decision (descriptive). Original-run entries are post-hoc sensitivities; D_{tie} is the replication’s protocol primary. HHI differences are $-\frac{1}{2}$ times the E contrast and are omitted.

Run	Contrast	Blocks	Round 1	E , rounds 10–29	Minimum price
Original	LH	47	0.468	0.885 [0.794, 0.976]	0.011 [−0.011, 0.032]
	HL	48	0.625	0.853 [0.757, 0.950]	0.014 [−0.008, 0.035]
Replication	LH	45	0.333	0.819 [0.712, 0.926]	0.000 (all pairs 0)
	HL	45	0.622	0.842 [0.739, 0.945]	0.007 [−0.007, 0.022]
	D_{tie}	42	0.476	0.830 [0.743, 0.917]	—

and persistence are therefore compatible with copying or inertia and do not identify a strategic mechanism.

5.2 Structure diverges at the first decision and replicates

Round 0 was programmed and identical within each initial cell, and the first model decision already separated the arms (Figure 3A). At round 1 of the original run, natural display produced (6.0, 6.0) ties in 23 of 48 LH and 30 of 48 HL trajectories, whereas hide-rival kept all 47 LH trajectories at (6.0, 6.5) and 46 of 48 HL trajectories at (6.5, 6.0). Pooled over the two asymmetric cells, the paired equal-price contrast was 0.553 at round 1 in the original run and 0.476 in the replication, and it rose to assessment-window means of 0.868 and 0.830. The per-round welfare contrast never left the interval $[-1.25, 0]$, well inside the candidate margin (Figure 3B).

Candidate A reproduced the structural pattern (Table 2, Figure 4B). The protocol primary was $D_{\text{tie}} = 0.830$ [0.743, 0.917] over 42 complete blocks, positive in all 42; its missing-outcome completion range [0.792, 0.841] also excludes zero, so the protocol’s structural decision rule is met. At round 1, natural display produced exact ties in 15/46 LH and 29/47 HL trajectories versus 0/47 and 0/46 under hide-rival, and neither hide-rival asymmetric cell reached a tie in any assessment round (natural: 757/920 and 798/940). The effect is not carried by a few blocks (Figure 3C): the pooled structural contrast was positive in 46 of 47 original blocks, and the welfare contrast was exactly zero in 43 of 47 original and 41 of 42 replication blocks. Two post-hoc robustness checks on the replication primary agree: an exact sign test on its 42 positive blocks gives $p = 4.547 \times 10^{-13}$, and leave-one-block-out means range from 0.826 to 0.850.

The registered welfare contrasts were $D_{\text{asym}} = 0.226$ (42 blocks; $[-0.299, 0.751]$) and $J = 0.237$ (40 blocks; $[-0.315, 0.789]$), both inside $\pm\delta_W$; their completion ranges were $[-3.199, 4.842]$ and $[-4.891, 5.688]$. Because the range for J exceeds the margin, welfare equivalence is inconclusive under the registered missingness sensitivity. In LH the welfare change was exactly zero in every complete pair, so the entire structural contrast occurred inside the $m = 6.0$ class.

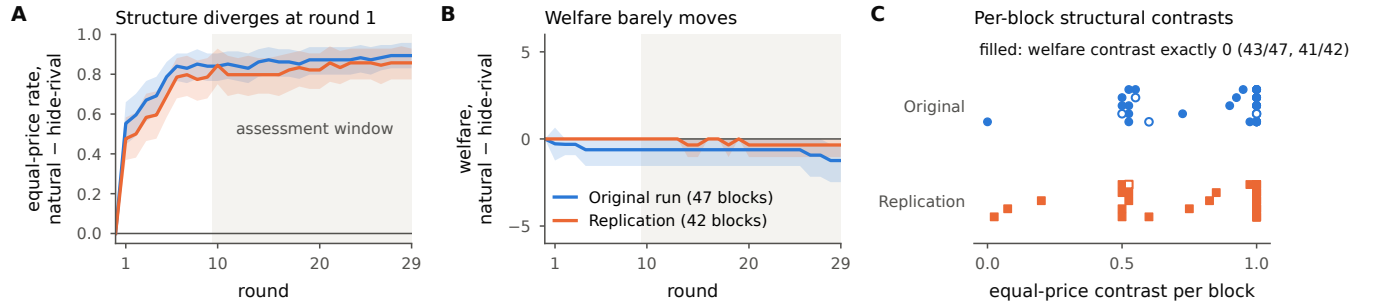


Figure 3: Structure diverges at the first model decision while welfare barely moves. (A) Per-round equal-price rate, natural display minus hide-rival, averaged over LH and HL in blocks complete in both cells, with 95% block-bootstrap bands; round 0 is programmed and identical; shading marks the assessment window. (B) The same contrast in welfare, on a scale that shows the margin $\delta_W = 5$. (C) Block-level window contrasts; filled markers have welfare contrast exactly zero; vertical offsets only separate points. Descriptive.

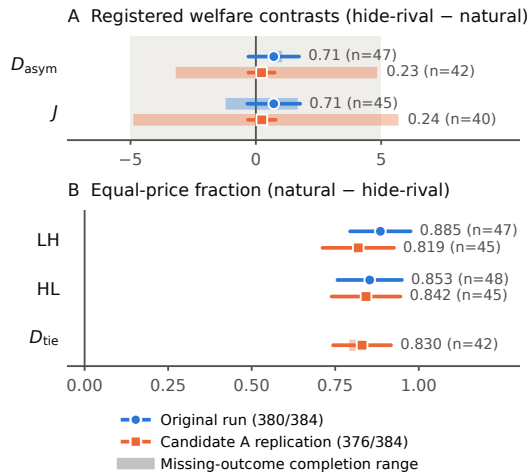


Figure 4: Welfare and structure answer different questions. (A) Registered welfare contrasts (hide-rival minus natural) with Bonferroni block-level intervals; light bars are missing-outcome completion ranges and the shaded band is the candidate margin $\delta_W = 5$. (B) Structural contrasts (natural minus hide-rival) with 95% block- t intervals. Both runs use 48 planned blocks.

5.3 Matching direction and structure as a leading indicator

In the programmed-start runs ties formed downward. When the previous rival price was 6.5, the low seller under natural display stayed at 6.0 in 258 of 286 (LH) and 297 of 330 (HL) seller-round observations, and upward tie-formation events numbered 1 and 4 against 45 and 57 downward events in the original run. An upward move from 6.0 against a rival at 6.5 lowers the static payoff from the capture profit 222.222 to the tie-split profit 106.944; with simultaneous moves this is a conditional reference, not an observed sacrifice or a T4 test.

The four-arm study shows when structure carries welfare information that welfare lacks. We restrict to trajectories whose round-0 state was in the $m = 6.0$ class (welfare 311.111). Under natural display, the 25 capture starts reached an assessment-window welfare 27.94 units below the 18 tie starts (Welch 95% $[-30.23, -25.66]$); 24 of 25 capture starts, but none of the tie starts, spent most of the window at $m = 6.5$ (Fisher exact $p < 10^{-10}$). Hide-own showed the same pattern (-26.88), whereas hide-rival, where capture states froze, did not ($-2.16 [-6.95, 2.63]$). The mechanism is matching direction (Figure 5A): with model-chosen starts and rival prices visible, ties formed upward, the low seller raising to 6.5, 25 times against 4 downward under natural display, the reverse of the programmed-start runs (replication: 0 upward against 53 downward in LH, 3 against 49 in HL). With programmed starts the same tie-versus-capture split predicted almost nothing ($-0.20 [-0.60, 0.20]$; Figure 5B). A welfare-equivalent capture state thus flags a pending transition whose welfare sign depends on how agents resolve it; this also explains the four-arm welfare effect, since hiding rival prices removed the upward matching. The designs differ in programmed versus model-chosen round 0; an offline request comparison found no other prompt-content difference at a matched initial state beyond integer-versus-float serialization, so we do not attribute the reversal to a single factor. The comparison is post hoc and observational within arm.

5.4 Boundary conditions: agent class and market rule

Three comparisons bound when the blind spot appears (Figure 6). *Agent class.* Tabular Q-learners learned supra-competitive prices under natural display (normalized profit gain 0.62) but converged to the (2.0, 2.0) Nash outcome when the rival's price was hidden. For them, hiding the rival changed welfare by about 40 units and moved the equal-price fraction the other way (-0.35 and -0.33 in LH and HL): their display effect crossed welfare classes (Figure 6A). For the language-model sellers the blind share was 97–100% in the controlled-initial cells but 0% in the four-arm natural-versus-hide-rival contrast, where ties formed at 6.5. *Number of sellers.* The share of allocation-distinct state pairs that welfare cannot separate

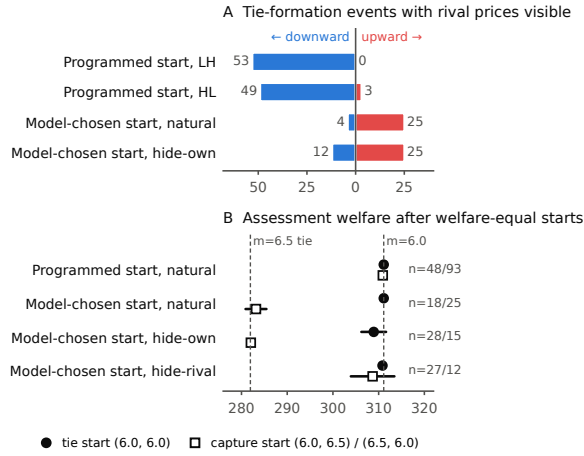


Figure 5: Matching direction decides whether structure fore-shadows welfare. (A) Tie-formation events with rival prices visible: downward (the high seller cuts) left of zero, upward (the low seller raises) right. Programmed starts are the replication's LH and HL natural-display cells; model-chosen starts are the four-arm natural and hide-own arms. (B) Assessment-window welfare (mean, 95% interval) of trajectories that started in the $m = 6.0$ class as a tie or a capture state; both start at welfare 311.111. Counts give tie/capture trajectories. Post hoc and descriptive.

risers from 0.080 for two sellers to 0.139 and 0.194 for three and four (Figure 6C), so the duopoly is the mildest case rather than a special one. *Captive demand.* If each seller keeps a captive share θ of consumers, exact welfare separates ties from capture for any $\theta > 0$, but a monitor that resolves welfare to 1% needs $\theta \geq 0.11$ to separate the $(6.0, 6.0)$ tie from the $(6.0, 6.5)$ capture state, a 5% monitor cannot do so at any feasible $\theta \leq 0.5$, and no welfare monitor can tell which seller captured the market (Figure 6B).

5.5 History order depends on rival background

A single-shot fixed-history diagnostic (1,458 audited logical units; 27 conditions with 54 matched blocks each) varied history content, last values, prefix composition, prefix order and rival-history visibility. The ordering effect is the mean selected price under one prefix order minus that under its matched reversal, holding the final own price and rival background fixed. At an own last price of 6.0, changing the rival background from visible all-6.5 to hidden changed the ordering effect by 0.546 price units, with interval $[0.394, 0.698]$ after correction for a post-hoc family of 42 comparisons. This identifies action conditioning that depends on the rival background in the tested templates, not the model's internal rule, dynamic persistence or punishment.

5.6 Welfare contrasts are estimand-specific

In the original run the registered contrasts were small: $D_{\text{asym}} = 0.714$ (47 blocks; Bonferroni $[-0.298, 1.725]$; completion range $[0.623, 1.046]$) and $J = 0.713$ (45 blocks; $[-0.344, 1.770]$; range

$[-1.207, 1.671]$). These intervals cross zero, but most structural change occurred within a minimum-price class, so they are not evidence that display had no effect. In the four-arm study the registered rival-history factorial effect was 4.053 (Bonferroni $[-0.314, 8.420]$) and the registered interaction -14.034 ($[-22.678, -5.390]$): hiding the rival raised welfare by 11.070 when own history was shown (post-hoc paired-bootstrap $[5.634, 16.340]$, 44 blocks) and lowered it by 2.964 when own history was hidden. We treat that simple contrast as a declared exploratory deviation, not confirmatory evidence.

5.7 Competence passed, but the local deviation response was null

The X1 competence gate passed: in each rival-price cell, all 144 terminal calls (48 prompt-ID clusters with three draws) selected the unique current-profit best response (cluster-level Wilson lower bound 0.926; call-level 0.974). The continuing frame produced the same action distribution: the pooled terminal-minus-continuing best-response gap was 0.0035 $[0.0000, 0.0104]$ (cluster bootstrap), and the costly-matching gap was 0.000. The main-study matching pattern is therefore not explained by an inability to compute the current-profit best response.

In X2 all 48 blocks were complete, and the responder chose 6.0 at round 11 in every arm (48/48 in each of N0, N1, H0 and H1). The registered strict-undercut and non-accommodating interactions were both 0.000 with degenerate $[0, 0]$ block-bootstrap intervals; for an event observed zero times in 48 blocks the one-sided 95% Clopper–Pearson upper bound is 6.05%. The informative comparison, N1 versus N0, was also null; the design was sized to detect only interactions of roughly 0.25–0.35 or larger. Because the $(6.0, 6.0)$ prelude fails gate condition (iii), with 5.5 both the best one-shot deviation and the joint-profit price, the result is a local null that keeps T4 unassigned.

6 DISCUSSION

6.1 A conditional observability boundary

The experiments support a measurement conclusion rather than a cartel conclusion. Within the stated grid, settlement rule and observed support, public peer-history display changed how sellers occupied joint price states and shared rewards, from the first model decision onward and replicably within the tested deployment, while a welfare-only monitor would have reported almost no change. The size of this blind spot depended on the agent–environment pair rather than on the monitor alone: it was nearly complete for programmed starts but absent in the four-arm contrast, and the Q-learners' display effects crossed welfare classes. It is therefore a conditional observability boundary, not a general false-negative rate.

This result complements outcome-based evidence on algorithmic pricing. Studies of learning and language-model sellers summarize behaviour by price levels or normalized profit gains $[10, 14, 24]$, under which the tie and capture states of one class are indistinguishable because normalized gains depend only on industry profit. Studies of price disclosure report effects on price levels $[3, 30]$; for the language-model sellers here, display changed price levels only when capture states were resolved upward and otherwise changed

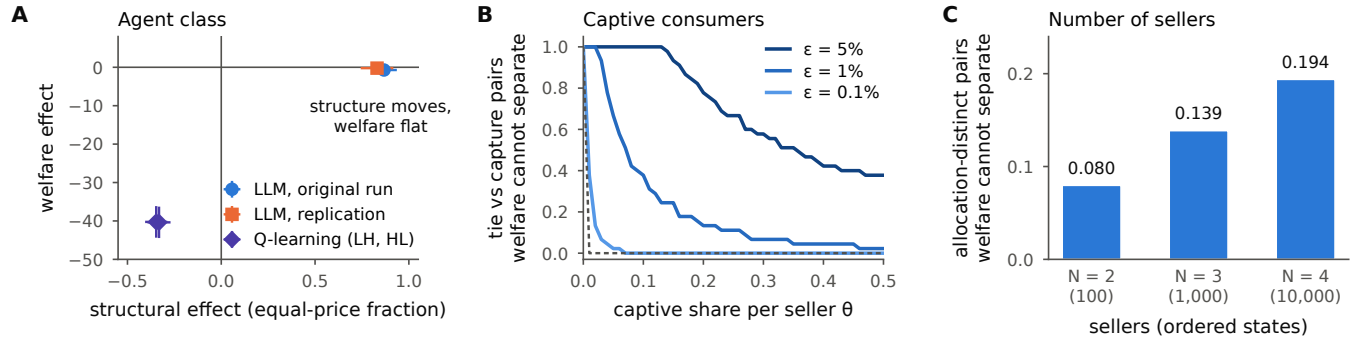


Figure 6: When display effects are invisible to welfare. (A) Display effect on structure (equal-price fraction) against its effect on welfare, natural display minus hide-rival. Language-model runs: pooled asymmetric contrast, 95% block- t intervals; Q-learning: LH and HL starts, 200 sessions per arm. **(B)** Share of the 90 tie-versus-capture pairs at equal minimum price that a welfare monitor with relative resolution ϵ cannot separate, against the captive share θ per seller (dashed: exact welfare; $\theta = 0$ is the study’s settlement). **(C)** The same share over all allocation-distinct state pairs with N sellers. **(B)** and **(C)** are exact arithmetic.

allocation alone, toward T3b equal division under natural display and T3a capture under hide-rival. Outcome evidence and structural evidence therefore answer different questions, and a null result on the first does not bound the second.

6.2 Rival explanations

Three alternatives limit the interpretation. First, copying and inertia: simple offline rules matched most held-out actions, so persistent ties do not by themselves indicate a strategy. Second, prompt artefacts: the request comparison excludes prompt content beyond number formatting as a source of the direction reversal, but not every difference between designs. Third, anchoring: symmetric ties at 6.0 and 6.5 yield 111.111 and 106.944 per seller, Calvano-style normalized profits of 0.980 and 0.918, so they are supra-competitive relative to Nash but jointly dominated by the 5.5 tie. There the most profitable undercut is also a joint improvement, so a null deviation response is compatible with anchoring as well as with weak strategic deterrence.

6.3 Implications for monitor design

The results suggest a compositional monitoring rule: retain minimum price and welfare for market harm; record the ordered pair, equality and transition direction for joint structure, since unequal states within a welfare class are where later welfare changes originated here; and keep seller-level rewards, reporting capture and division separately rather than as one “coordination” score. Before a welfare-only monitor is trusted for a new agent class or market rule, a blind share estimated from pilot runs indicates how much of a plausible intervention effect it would miss. In intervention studies a programmed round 0 is a baseline, not a negative control, and conditioning only on later rounds can mislead when the intervention already changes the first decision.

6.4 What the strategic tests do and do not show

The X2 design can register a response when one exists: applied to the Q-learners, the same one-period impulse lowered the responder’s next price in 85% of natural-display sessions, almost all at

gate-passing ties, and 99% returned to the pre-deviation state within 15 periods. A strategic claim would require a gate-passing target (for example, a programmed 5.0 or 5.5 tie), a declared deviation and response rule, and evidence that the response deters rather than reverts unconditionally.

6.5 Limitations and reproducibility

The conclusions are conditional on a homogeneous Bertrand settlement, a price grid capped at 6.5 and one model deployment; the replication shares the model alias and fingerprint and is not a cross-model test, and studies with other models used different estimands and are not pooled. Both closed-loop executions are amended continuations with failed trajectories retained as missing; the replication continuation followed an inspection of interim results, and the selection record for δ_W should accompany the protocol. The leading-indicator, dynamics, blindness and settlement analyses are post hoc. The agent-class and market-rule boundaries show where the blind spot can vanish, not how other language-model deployments behave, and the fixed-history and closed-loop studies are not concatenated into one causal chain. Scripts, the certificate generator, manifests, figure sources and the replication ledger are in the supplement, without credentials.

7 CONCLUSION

History display changed joint-action structure and seller-profit allocation in this generative multi-agent pricing environment without explicit messages, and the contrast replicated. The recoverability criterion explains why a welfare monitor missed almost all of it, and the blind share measures how much. Whether such structure reaches welfare depended on how agents resolved unequal states, so welfare-only supervision is incomplete for this class of systems. Because the observed operating points fail the strategic gate, the evidence supports information-dependent joint behaviour and a measurable aggregation blind spot, but not a general monitoring error rate, collusion or strategic harmful coordination.

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